motions. At zero gravity, the mere start and stop of the on-board tape recorder can jiggle the spacecraft enough to smear the picture.

This problem was solved by sending up commands to the spacecraft's little rocket engines (called thrusters), machines of exquisite sensitivity. With a little puff of gas at the start and stop of each data-taking sequence, the thrusters compensated for the tape-recorder jiggle by turning the entire spacecraft just a little. To deal with the low radio power received at Earth, the engineers devised a new and more efficient way to record and transmit the data, and the radio telescopes on Earth were electronically linked together with others to increase their sensitivity. Overall, the imaging system worked, by many criteria, better at Uranus and Neptune than it did at Saturn or even at Jupiter.

Voyager may not yet be done exploring. There is, of course, a chance that some vital subsystem will fail tomorrow, but as far as the radioactive decay of the plutonium power source is concerned, the two *Voyager* spacecraft should be able to return data to Earth roughly through the year 2015.

Voyager is an intelligent being—part robot, part human. It extends the human senses to far-off worlds. For simple tasks and short-term problems, it relies on its own intelligence; but for more complex tasks and longer-term problems, it turns to the collective intelligence and experience of the JPL engineers. This trend is sure to grow. The *Voyagers* embody the technology of the early 1970s; if spacecraft were designed for such a mission today, they would incorporate stunning advances in artificial intelligence, in miniaturization, in data-processing speed, in the ability to self-diagnose and repair, and in the propensity to learn from experience. They would also be much cheaper.

In the many environments too dangerous for people, on Earth as well as in space, the future belongs to robot-human partnerships that will recognize the two *Voyagers* as antecedents and pioneers. For nuclear accidents, mine disasters, undersea exploration and archaeology, manufacturing, prowling the interiors of volcanos, and household help, to name only a few potential applications, it could make an enormous difference to have a ready corps of smart,